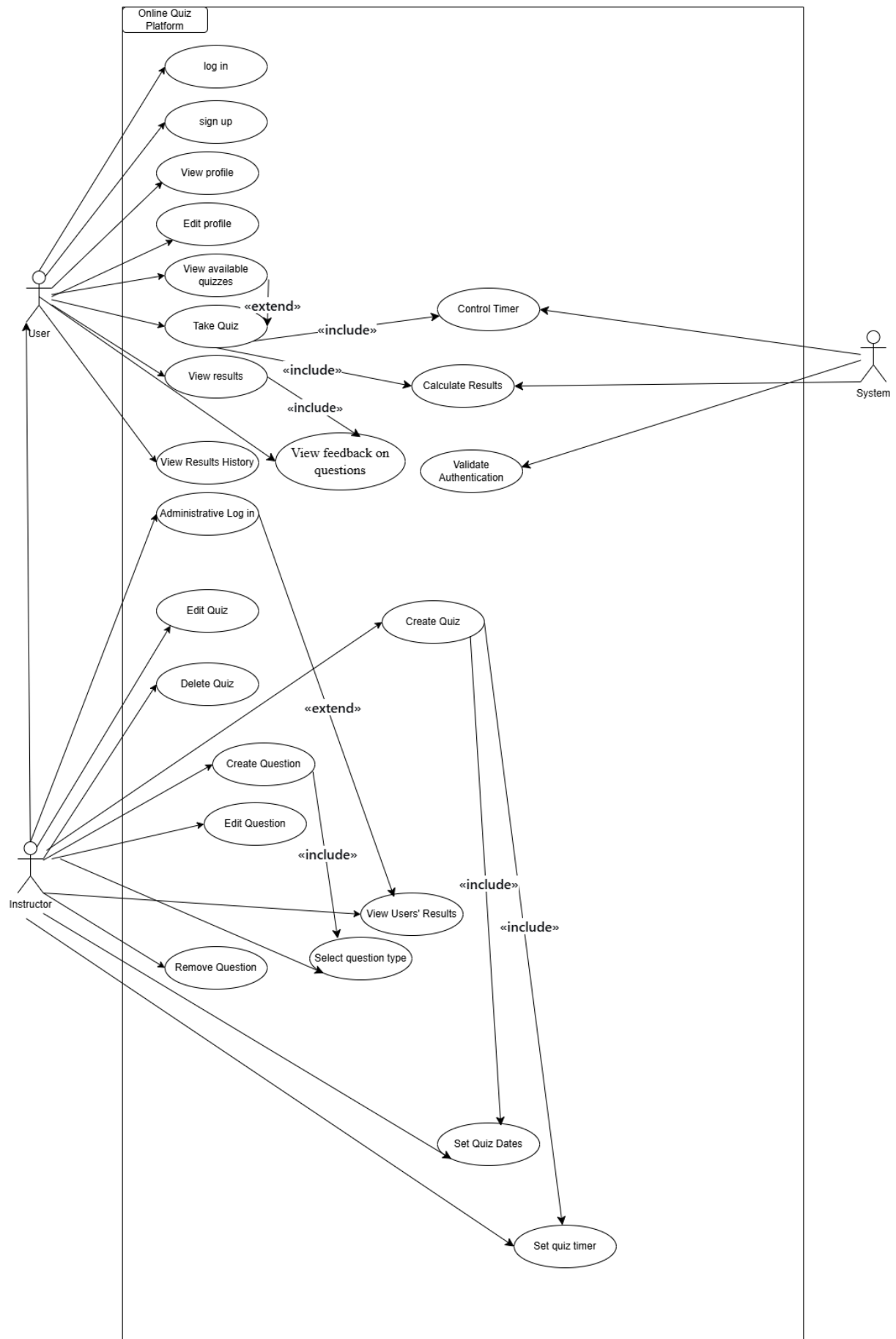




Use Case Diagram

1. Description Table

Actor	Use case	Description
User	Log in	Login using a username/valid email address and password.
User	Sign Up	Register by providing username, valid email address, and password.
User	View Profile	View personal data.
User	Edit Profile	Edit personal data.
User	View available quizzes	View a list of available quizzes.
User	Take quiz	Select a quiz and answer the questions.
User	View Result	View quiz result.
User	View Results history	View history of completed quizzes and scores.
User	View feedback on questions	View a summary at the end of the quiz indicating which answers the user got right and which ones got wrong.
Instructor	Administrative Log in	Log in with administrative credentials, ensuring exclusive access to the management panel.
Instructor	Edit Quiz	The instructor can edit a quiz.
Instructor	Delete Quiz	The instructor can delete a quiz.
Instructor	Create Quiz	The instructor can create a quiz.
Instructor	Create Question	The instructor can create a question.
Instructor	Edit Question	The instructor can edit a question.
Instructor	Remove Question	The instructor can remove a question.
Instructor	Select Question Type	The instructor can select a question type.
Instructor	Set Quiz Dates	The instructor can set a date for the quiz.
Instructor	Set Quiz Timer	The instructor can set a timer for the quiz.
Instructor	View Users' result	The instructor can see the users' quiz results.
System	Control Timer	The system can control the timer for the quiz.
System	Calculate results	The system calculates the result after a quiz.
System	Validate Authentication	The system validates instructor/user login authentication.

2. Relation Table

Relation	Description
Generalization: Instructor -> User	Instructor inherits all the user's functionality
Take quiz <<include>> Control timer	While taking a quiz, the system controls the time.
Take Quiz <<include>> Calculate Results	When finishing a quiz, the system calculates the score.
View results <<include>> View feedback on questions	When viewing results, the system displays detailed feedback on the quiz questions.

Create quiz <<include>> Create Question	Creating a quiz may include creating questions.
Create quiz <<include>> Set Quiz dates	Creating a quiz includes setting dates.
Create quiz <<include>> Set quiz timer	Creating a quiz includes defining the total time.
Create Question << include>> Select Question Type	Creating a question includes selecting the type.
Administrative Login <<extend>> View Users' Result	After administrative login, the instructor can optionally view user results.
View Available <<extend>> Take Quiz	When viewing available quizzes, the user can optionally start a quiz.